

40.520 Stochastic Models

Brownian Motion

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Outline

1. Definition of Brownian Motion
2. Hitting Times and Maximum Variable
3. Moment Generating Function
4. Drifted Brownian Motion

Definition of Brownian Motion

Symmetric Random Walk

- The symmetric random walk can be represented by

$$Z_n = \sum_{i=1}^n X_i$$

with $\{X_i, i \geq 1\}$ being a sequence of independent random variables, whose distribution is given by

$$P(X_i = 1) = P(X_i = -1) = 1/2$$

- The symmetric random walk is not only a **Markov process** but also a **Martingale process**.

Independent and Stationary Increments

Take a sequence of integers:

$$0 = i_0 < i_1 < \dots < i_n.$$

- Independent increments:

$$Z_{i_1} - Z_{i_0}, Z_{i_2} - Z_{i_1}, \dots, Z_{i_n} - Z_{i_{n-1}}$$

are independent.

- Stationary increments: the distribution of

$$Z_{n+m} - Z_n$$

does not depend on n .

Scaled Random Walk

- Rescale time and space for the random walk model. That is, let

$$Y_t = \Delta x \cdot \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i.$$

- **Mean:**

$$E[Y_t] = 0$$

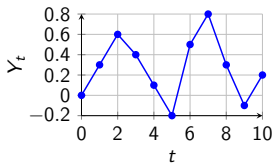
- **Variance:**

$$\text{Var}[Y_t] = (\Delta x)^2 \lfloor \frac{t}{\Delta t} \rfloor$$

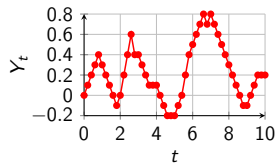
Limits of Scaled Random Walk

- We now let Δx and Δt go to 0 in such a way that

$$\Delta x = \sqrt{\Delta t}.$$



Unscaled Random Walk ($\Delta t = 1$)



Scaled Random Walk ($\Delta t = 0.2$)

Limits of Scaled Random Walk

- By the central limit theorem,

$$Y_t \sim N(0, \sqrt{t}) \text{ as } \Delta x \rightarrow 0.$$

Brownian Motion

A stochastic process $\{X_t, t \geq 0\}$ is said to be a (standard) Brownian motion process if:

- $X_0 = 0$;
- It has stationary and independent increments;
- For every $t > 0$, X_t is normally distributed with mean 0 and variance t ;
- It has continuous sample paths.

Brownian Motion as Markov and Martingale Processes

- The independent increment assumption implies that Brownian motion is a Markov process:

$$P\{X_{t+s} \leq a \mid X_u, 0 \leq u \leq s\} = \Phi \left[\frac{a - X_s}{\sqrt{t}} \right],$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

- It also implies that Brownian motion is a martingale (continuous-time):

$$E[X_{t+s} \mid X_u, 0 \leq u \leq s] = X_s.$$

Covariance of Brownian Motion

Covariance: if $s < t$,

$$\text{Cov}[X_s, X_t] = \text{Var}(X_s) = s.$$

Example

- For $t_1 < t_2 < t_3$, compute

$$E[X_1 \cdot X_2 \cdot X_3].$$

Hitting Times and Maximum Variable

Distribution of Hitting Time

Let T_a be the first time the Brownian motion process hits a level $a > 0$. Then

$$P\{T_a \leq t\} = 2P\{X_t \geq a\} = \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^{+\infty} e^{-\frac{y^2}{2}} dy.$$

which implies that the probability density function of T_a is

$$f_{T_a}(t) = \frac{a}{\sqrt{2\pi}} t^{-\frac{3}{2}} e^{-\frac{a^2}{2t}}.$$

Maximum Value of a Brownian Motion

- For $a > 0$,

$$P\left\{\max_{0 \leq s \leq t} X_s \geq a\right\} = P\{T_a \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^{+\infty} e^{-\frac{y^2}{2}} dy.$$

Example: Hitting Probability of Brownian Motion

- Note that Brownian motion is a martingale process. We still can use the martingale stopping theorem to compute

$$P(X_T = A) = \frac{B}{A + B}$$

where T is the first time the Brownian motion X hits A or $-B$.

Moment Generating Function

Moment Generating Function (MGF)

Definition: For a random variable X , the moment generating function

$$M_X(t) = \mathbb{E} \left[e^{tX} \right],$$

where t is a real number.

MGF: Examples

Each distribution has a unique MGF. For example:

Distribution	PDF / PMF	MGF $M_X(t)$
Normal $N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Exponential (λ)	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{\lambda}{\lambda - t}, t < \lambda$
Poisson (λ)	$e^{-\lambda} \frac{\lambda^k}{k!}$	$e^{\lambda(e^t - 1)}$
Binomial (n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$	$(1 - p + pe^t)^n$

Key Properties

1. Sum of Independent RVs:

If $Z = X + Y$ with $X \perp Y$, then $M_Z(t) = M_X(t) \cdot M_Y(t)$

2. Linear Transformation:

$$M_{aX+b}(t) = e^{bt} M_X(at)$$

3. Uniqueness:

If $M_X(t) = M_Y(t)$ for all t , then X and Y have the same distribution.

4. Moment Extraction:

$$\mathbb{E}[X^n] = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}$$

Example: Sum of Independent Normals

Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, independent. Show

$$Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Brownian Motion with Drift

Brownian Motion with Drift

Standard Brownian Motion B_t :

- No trend: $\mathbb{E}[B_t] = 0$
- Variance grows linearly: $\text{Var}(B_t) = t$

But real-world processes have:

Process = Deterministic Trend + Random Noise

Brownian Motion with Drift

- Let $\{B_t, t \geq 0\}$ be a standard Brownian motion process. If we attach it with a deterministic drift, i.e., let

$$X_t = \sigma B_t + \mu t,$$

we say that $\{X_t, t \geq 0\}$ is a Brownian motion with drift μ and volatility σ .

- **Mean:** $E[X_t] = \mu t$
- **Variance:** $Var[X_t] = \sigma^2 t$

Drifted Brownian Motion

The above drifted Brownian motion has the following properties:

- $X_0 = 0$;
- It has stationary and independent increments;
- X_t is normally distributed with mean μt and variance $\sigma^2 t$.

Geometric Brownian Motion

If $\{X_t, t \geq 0\}$ is a Brownian motion process with drift μ and volatility σ , then the process $\{Y_t, t \geq 0\}$ defined by

$$Y_t = e^{X_t}$$

is called a geometric Brownian motion.

- Positivity
- Proportional random changes (returns)
- Matches real data (lognormal distribution)

Example

Let $Y_t = e^{X_t}$, where $\{X_t, t \geq 0\}$ is a Brownian motion process with drift μ and volatility σ . Show

$$E[Y_t | Y_s = y] = y \cdot e^{(t-s)(\mu + \sigma^2/2)}$$